Classical apparatus: construction, complete records and stable recovery

A finite autonomous Hamiltonian apparatus can reconstruct a known spring receiver after a fixed latency with arbitrarily small error and disturbance. At fixed nonzero coupling, exact nonlinear calibration instead permits vanishing reconstruction error with bounded disturbance. Complete final clock records replace initial clock calibration, provided incoming probe momenta are known. These are different limits of one apparatus; none derives a positive universal action. The complementary preparation obstructions and calibration thresholds are proved in the calibration treatment.

This treatment consolidates C066–C069, C088–C089, C094–C105 and the construction lemmas used by C090–C123. Historical coordinator/model checks were internal written checks, not external peer review; novelty is unassessed. Source reading levels and the original internal checks remain in the linked B35, B37, B47 and B50–B55 records. All estimates below are written derivations.

Receiver and one autonomous apparatus

The reduced three-body receiver of the receiver treatment has state $z = (x, P, y, Q)$ and

$$H_s = \frac{P^2}{2\mu} + \frac{Q^2}{2\nu} + \frac{a}{2}x^2 - gxy + \frac{d}{2}y^2, \qquad (\mu, \nu, a, g, d) = (3m/2, m/2, 9k/4, 3k/4, 5k/4).$$

Here $m, k > 0$, $ad - g^2 > 0$, and P is canonical; physical tagged momentum is $2P/3$. Write $\Phi_t = e^{Mt}$ for its free flow and G for its positive energy matrix, $H_s = z^T G z/2$. Fix $E, T > 0$ and a compact convex receiver neighbourhood Z containing $\{H_s \le E\}$, with an open enlargement for derivatives. Apparatus preparation and all norms use fixed positive component length/momentum units, including receiver units L_*, P_*. Constants may depend on these units, the design and T; they are not uniform as $T \to 0$.

Add a clock (s, p_s) of mass $M_c > 0$ and four probes (q_j, π_j) of masses $M_j > 0$. With fixed stiffness $K > 0$ and dimensionless coupling $0 < \lambda \le \lambda_0$, take

$$H_\lambda = H_s + \frac{p_s^2}{2M_c} + \sum_{j=1}^{4} \frac{\pi_j^2}{2M_j} + \lambda K \sum_{j=1}^{4} f_j(s) X(x) R(q_j).$$

X, R are bounded smooth coordinate cutoffs, equal to their arguments on fixed trajectory neighbourhoods. Smooth nonnegative compactly supported f_j have disjoint supports strictly inside the nominal clock path $s_0 + v_0 t$, $0 < t < T$, $v_0 > 0$, with fixed endpoint margins. All five masses, pulse shapes, widths, derivatives and T are fixed before any coupling or precision limit. Kinetic energy is positive; a fixed additive constant makes the bounded interaction plus Hamiltonian nonnegative. The finite-dimensional potential assumes neither spatial locality nor finite signal speed.

Energy conservation bounds receiver coordinates and momenta on the admitted energy class. Bounded cutoffs bound interaction forces globally. Together with compact initial apparatus boxes these give uniform force and speed ceilings over the fixed horizon; harmonic forces have no asserted global ceiling outside that energy class. Reducing λ_0 after design meets any positive apparatus-force ceiling and preserves cutoff margins, pulse traversal and positive clock momentum. Persistent momenta are stored after the clock exits all supports. Position records are at the known time T; unlimited bounded storage space is not asserted.

Four signals and the common smooth-flow estimates

The free output's first four derivative rows are

$$x = x_0, \quad \dot{x} = P_0/\mu, \quad \ddot{x} = (-ax_0 + gy_0)/\mu, \quad x^{(3)} = -aP_0/\mu^2 + gQ_0/(\mu\nu).$$

They form an invertible triangular observation matrix, with diagonal $1, 1/\mu, g/\mu, g/(\mu\nu)$. Analyticity implies that the evaluation rows $e_x\Phi_t$ span four dimensions on any open time interval. Otherwise an annihilated vector would have identically zero output and all four initial coordinates zero. Select four independent evaluation times and then sufficiently narrow, positive, fixed pulse widths. Row-normalized pulse integrals approach these evaluation rows, so

$$A_{c,j\cdot} = K \int_0^T f_j(s + vt) e_x \Phi_t \, dt, \qquad c = (s, v),$$

are invertible. In a small fixed clock rectangle a uniform inverse margin $\|A_c h\| \geq \alpha \|h\|$, $\alpha > 0$, follows by continuity. $A_c z$ has momentum units before component conversion.

In the linear cutoff regions the exact equations and integrals include

$$\dot{\pi}_j = -\lambda K f_j(s) x, \quad \dot{P} = -ax + gy - \lambda K \sum_j f_j(s) q_j, \quad \dot{p}_s = -\lambda K \sum_j f_j'(s) x q_j,$$

$$\pi_j(T) - u_j = -\lambda K \int_0^T f_j(s(t)) x(t) dt,$$

$$q_j(T) - q_j - T u_j / M_j = -\frac{\lambda K}{M_j} \int_0^T (T - t) f_j(s(t)) x(t) dt,$$

$$p_s(T) - M_c v = -\lambda K \sum_j \int_0^T f_j'(s(t)) x(t) q_j(t) dt,$$

$$s(T) - s - vT = -\frac{\lambda K}{M_c} \sum_j \int_0^T (T - t) f_j'(s(t)) x(t) q_j(t) dt.$$

Primes on pulses mean derivatives with respect to clock position. On a fixed compact preparation domain, variation of constants and the smooth finite-time variational equations bound coupled minus free trajectories by $O(\lambda)$ in C^1 in every initial variable. The same is true of the integrands and their initial derivatives. Dividing the momentum integral by λ gives a smooth extension at zero coupling. Thus every scaled map below has a uniform $O_{C^1}(\lambda)$ remainder on a convex enlarged domain. Clock reaction is included, rather than replaced by an external clock. Smoothness also holds for a small signed coupling interval used only for Taylor proofs; physical coupling is positive.

Two reusable estimates organize all the recovery statements. If $F = L + R$ on a convex domain, L has a global inverse margin $\beta > 0$, and $\|DR\| \leq \beta/2$, segment integration gives

$$\|F(w) - F(w')\| \geq \tfrac{\beta}{2} \|w - w'\|.$$

For a compact physical subset, a minimum-residual fit to data with error δ exists. Its exact output and the true output differ by at most 2δ, hence

$$\|\widehat{w} - w\| \leq 4\delta/\beta, \qquad \epsilon_x \epsilon_P \leq 16 L_* P_* \delta^2/\beta^2.$$

These are sufficient stability bounds, not exact positive-error minimax constants. Segments need only stay in the enlarged convex domain, not on an energy shell. A pointwise rank calculation alone does not give this result.

Quiet preparation and fixed-coupling calibration

First prepare all ten apparatus coordinates in a full rectangular box of half-width $b > 0$ about zero probe phase and $(s_0, M_c v_0)$. Each box has positive phase volume. Probe force integration gives $q, \pi = O(b + \lambda)$; receiver forcing then gives

$$z(t) - \Phi_t z_0 = O(\lambda b + \lambda^2), \qquad (s(t), p_s(t)) - (s_0 + v_0 t, M_c v_0) = O(b + \lambda^2).$$

Finite-time continuation justifies the linear cutoff equations uniformly. Inserting these estimates in the momentum integrals gives

$$\boldsymbol{\pi}(T) = \boldsymbol{\pi}(0) - \lambda A z_0 + O(\lambda b + \lambda^3).$$

Consequently the first-order inverse $-A^{-1}\boldsymbol{\pi}(T)/\lambda$ has error $O(b/\lambda + b + \lambda^2)$. Taking $b = \lambda^3$ makes both reconstruction and receiver disturbance $O(\lambda^2)$, with exact final momentum access. With final momentum error ρ, add $O(\rho/\lambda)$; $\rho = O(\lambda^3)$ retains the rate. For

$$\mathcal{U} = \epsilon_x \sup_{0 \le t \le T} |P(t) - P^0(t)|,$$

one obtains $0 \le \mathcal{U} \le C L_* P_* \lambda^4 \to 0$. Clock mean energy, apparatus mass and duration stay fixed; incoming phase volume and record errors shrink. Reporting $\Phi_{t_i} \widehat{z}_0$ on any partition uses these same four records after latency T. This does not supply fresh causal measurements at every cut or recover an unknown force.

At nominal apparatus data the probes are $O(\lambda)$, and receiver/clock reaction is $O(\lambda^2)$. Differentiating their equations in z_0 preserves these orders: reaction derivatives contain λ times a probe or its variation. The nominal nonlinear four-momentum map therefore obeys

$$F_\lambda(z) = -\lambda A z + \mathcal{R}_\lambda(z), \qquad \|\mathcal{R}_\lambda\|_{C^1} \le C\lambda^3.$$

For small positive coupling its lower Lipschitz margin is $\lambda\alpha/2$. Fix one such coupling. Apparatus uncertainty b and record error ρ perturb the nominal record by at most $C_\lambda b + \rho$. Minimum-residual calibration on $H_s = E$ gives

$$\|\widehat{z} - z\| \le \frac{4}{\lambda\alpha}(C_\lambda b + \rho).$$

Thus exact nonlinear calibration removes the first-order inverse's fixed $O(\lambda^2)$ systematic remainder. At fixed coupling, shrinking b, ρ closes both reconstruction-coordinate product and $\mathcal{U}$, while receiver disturbance need only stay bounded. Exact model knowledge is supplied; no computational-efficiency or uncertain-model theorem is claimed. The record perturbations are contained in a small norm ball, without proving that every point of an independently specified oracle-error ball is realized.

Complete records replace clock preparation data

Known incoming probe momenta are denoted u. When the clock data c are known, subtract the known free drift $d_j(u) = T u_j / M_j$ and use

$$F_{\lambda,c,u}(z, q) = ((\pi(T) - u)/\lambda, q(T) - d(u)) = (-A_c z, q) + O_{C^1}(\lambda).$$

The leading inverse margin is $\beta = \min(\alpha, 1)$. Eight final pointer coordinates recover z, q globally on the fixed bounded domain, with $\delta = \max(\rho_\pi/\lambda, \rho_q)$. Unknown incoming positions may have fixed positive widths. The bound above gives product $16 L_* P_* \delta^2/\beta^2$. The special position-response rank used in the calibration obstruction is unnecessary for this recovery theorem.

If initial offset is known but speed is hidden, set $u = 0$ and add final clock momentum. The leading map of the nine scaled records is

$$L(z, q, v) = (-A_v z, q, v).$$

Let $D = \sup \|(\partial_v A_v)z\|$ on the domain. For output distance d, its last blocks give $|\Delta v|, \|\Delta q\| \le d$, and the first gives $\alpha\|\Delta z\| \le (1 + D)d$. Hence the global inverse margin is $\beta = \min(1, \alpha/(1 + D))$. The $O_{C^1}(\lambda)$ remainder reduces it at most to $\beta/2$. Use $\delta = \max(\rho_\pi/\lambda, \rho_q, \rho_c)$ in the common bound. This recovers speed jointly with reaction; $p_s(T)/M_c$ is not substituted as an exact initial speed. Exact receiver energy and a local-patch prior are unnecessary.

If initial offset also is hidden, add final clock position at T. With clock units S_*, V_* and $\theta = TV_*/S_*$ the ten-record leading map is

$$L(z, q, s, v) = (-A_{s,v} z, q, s + \theta v, v).$$

Let D_s, D_v bound the signal's clock derivatives applied to receiver states. The inverse clock shear gives $|\Delta v| \le d$ and $|\Delta s| \le (1 + \theta)d$, so an inverse margin is

$$\beta = \min\{(1 + \theta)^{-1}, \alpha/[1 + D_s(1 + \theta) + D_v]\} > 0.$$

Again subtract the $C\lambda$ Lipschitz remainder and use the common bound, now with $\delta = \max(\rho_\pi/\lambda, \rho_q, \rho_s, \rho_c)$. All ten unknowns z, q, s, v are recovered on the whole bounded domain. These statements assume $u = 0$ is known (the eight-record statement allows known nonzero u); they do not assert recovery with completely unknown incoming apparatus phase. Simultaneous classical access to final positions and momenta is supplied, not implemented by a second recording apparatus.

Energy-assisted local recovery and its global obstruction

Without final clock records, a known initial offset and receiver energy can replace speed locally. At a selected shell point set $B = -A^{-1}\partial_v A$ and require $\tau = z_*^T G B z_* > 0$. Append H_s/E_* to the eight scaled pointer outputs. Its derivative kernel obeys $dq = 0$, $dz = Bz_* dv$, and $\tau dv/E_* = 0$, hence is zero. If this derivative is J, continuity gives one convex neighbourhood U and $\gamma = \|J^{-1}\|^{-1} > 0$ with $\sup_U \|D\mathcal{G}_\lambda - J\| \le \gamma/2$. Segment integration then gives the common recovery bound with $\beta = \gamma$ on a fixed shell patch times positive position/speed boxes inside U.

Existence of the transverse design and the actual distinct-speed global pairs are proved together in the clock-fibre construction. Its full-shell branches do not contradict this local result: a supplied patch is information. The inverse constants may deteriorate as $\tau \to 0$. Zero-coupling scaled extensions are proof devices; unscaled zero-coupling records contain no receiver signal.

Impulsive probes and actual repeated-cut refinement

A different instrument performs fresh readouts at each cut. Its externally switched ideal Hamiltonian impulses are $G_x = \alpha x \pi$ and $G_P = \beta P \rho$, with $\alpha > 0$ dimensionless, $[\beta] = T/M$ and canonical probes $(q, \pi), (r, \rho)$. Receiver/probe drift during impulses is omitted by definition. Exact symplectic shears give pre-cut estimates and kicks

$$\widehat{x} = x + q/\alpha, \quad \widehat{P} = P + r/\beta - \alpha\pi, \quad d_x = \beta\rho, \quad d_P = -\alpha\pi.$$

For full incoming rectangular half-widths (s_q, t_q, s_r, t_r), exact support errors are s_q/α, $s_r/\beta + \alpha t_q$, and disturbances are $\beta t_r, \alpha t_q$. Intrinsic products are $s_q t_q, s_r t_r$, in action units. Scaling all widths to zero through positive-volume classical preparations closes them; Liouville evolution

of a fixed preparation imposes no lower bound across different initial preparations. Exact final position records and switching/preparation resources are supplied. These internal receiver couplings are not local interactions with particle 1 alone.

For cut separation δ, block the receiver flow as $(A_\delta, B_\delta; C_\delta, D_\delta)$ in observed/hidden pairs $u = (x, P), w = (y, Q)$. With $\widehat{u}_i = u_i + e_i$, the exact inversion error is

$$\widehat{w}_i - w_i = B_\delta^{-1}(e_{i+1} - A_\delta e_i + A_\delta d_i).$$

The receiver proof gives

$$B_\delta^{-1} = \begin{pmatrix} 6\mu/(g\delta^2) + O(1) & -2/(g\delta) + O(\delta) \\ -12\mu\nu/(g\delta^3) + O(\delta^{-1}) & 6\nu/(g\delta^2) + O(1) \end{pmatrix}.$$

With fixed units and $\eta = \delta/t_*$, widths $O(\eta^p)$ therefore give hidden-state error $O(\eta^{p-3})$, including the intervening kick. At $N = T/\delta$ equal cuts the exact disturbance is $\sum_{j<i} \Phi_{(i-j)\delta}(d_j, 0)$, hence $O(\eta^{p-1})$. For $p = 4$ this gives uniform state reconstruction $O(\eta)$, disturbance and energy error $O(\eta^3)$, intrinsic products $O(\eta^8)$, and incoming free-probe kinetic-energy sum $O(\eta^7)$. The last sample is obtained by propagating the preceding hidden estimate, adding only $O(\eta^4)$ error. There is one sample interval of latency. Two fresh probes per cut require $2(N + 1) = O(\eta^{-1})$ probe count and fixed-probe-mass total mass. Reset, switching, trapping, storage and recording costs are not included. Finite meshes are different interventions with a common unmeasured-trajectory limit, not restrictions of one measured law. Fixed error widths make this inversion ill-conditioned, without proving impossibility for every estimator.

Source scope and consequence for STATE

The classical translation interaction is read in Katagiri v2 §5 (B35 companion); Theurel's thermal/conserved-momentum comparison is bounded by B36 and B37. The observability precedent is Hermann–Krener, routed through B34. B47 and B50–B55 document smooth-flow and inverse methods; the explicit matrices, reaction estimates and global margins here are model derivations. Historical checks cover specified claims, not a general external referee certification.

For STATE, delayed recovery of known dynamics, actual repeated interventions, and prediction under an unresolved force must remain separate. The latter has its own bounded-force theory. A physical restriction on preparation, complete-record access or calibration must be derived before any of these prepared information scales can exclude zero universally or identify a scale with physical $\hbar$.